Review and Reflection of the Agile Experience in the SNHU Travel Project

Brendan Malloy

Southern New Hampshire University

CS 250: Software Development Lifecycle

Joseph Martinez

October 18, 2026

Review and Reflection of the Agile Experience in the SNHU Travel Project

ChadaTech asked our team to pilot Scrum in developing the SNHU Travel booking application before making the decision to switch all teams from the waterfall process. During this project, I filled all roles. As the Product Owner, I drafted the user stories; as the Developer, I developed the Top Five Destinations list and created the wellness slide show; as the Tester, I wrote the test cases. In the Sprint Retrospective, I take the Scrum Master position and evaluate the factors that contributed to our success, prevented our failure, and the suitability of Scrum for the project.

Applying the Roles

Every role owned a particular decision, so that there were no delays related to figuring out who had to make which decision. As the president of SNHU Travel, Amanda set the business goal and the five-week deadline that became part of our team charter. As the Product Owner, Christy was responsible for ordering the Product Backlog. When I was in the Product Owner role, I converted the focus group meeting results into six backlog items, ordered them by priority, sized them to be either small, medium, or large, and composed detailed stories with acceptance criteria. These criteria defined the same concept of Done prior to any coding.

The Developers developed user stories into a product increment and verified that the resulting increment matched the acceptance criteria of its user story. As the Developer, I checked my Top Five Destinations list written in Java Swing against the acceptance criteria and saw that just a list itself would not satisfy the first one that required the user to click a link and access the list. I added a landing page with that link and called the story Done. The Tester, Brian, verified that the product increment satisfied the user story. As the Tester, I included even the behavior not mentioned in the stories, like pressing Next on the last slide. Ron, our Scrum Master throughout

the Sprints, organized Scrum events and resolved impediments. The charter rule stating that no one may go to Christy directly and hand to the Developer a development assignment outside the planned Sprint Backlog is what the Scrum Master exists for. Verwijs and Russo (2023) identified that the Scrum teams perform better in staying close to their stakeholders' needs and being able to respond to them, and the sharp differentiation of roles made it possible.

Completing the User Stories

With Scrum, we completed the user stories by breaking down the product to small pieces each having a clear finish line. Since the Product Backlog was ordered and sized, we developed the Top Five Destinations list first, before implementing the personalization feature, the filters and the hot deals. The acceptance criteria were then passed directly into testing. The Save Travel Preferences story requires the following items: a label on each control, an unset default state of the controls, and a confirmation message upon saving the control settings. All three criteria became pass or fail conditions for test case TC-01. The value statement was crucial as well. The returning traveler has to save his or her preferences so that the site matches the actual trips made. I concluded from it that saving preferences through the login session was expected, and I included this in my test cases. A story was considered completed when the tests passed.

Handling the Interrupts

There were two interruptions. The first happened after I asked Christy several questions concerning my test cases. She answered with a wireframe with one destination on one slide and Previous and Next buttons. The acceptance criteria remained unchanged, but user actions changed. So, I updated my three test cases. There was nothing built based on these tests at this stage, so this cost was limited to a rewriting effort on the paper.

The second interruption was much bigger – SNHU Travel management decided to focus the booking tool on the detox and wellness travel. Christy clarified that there was no intention to scrap the completed work and that only vacation types would change. Being the Developer, I preserved the functionality of the slide show navigation and replaced five slides with the wellness trips, e.g., the yoga and mindfulness retreat, the natural hot springs destination. Working with the code, I noticed that in the provided sample project both the caption and the button panels were placed in the same layout region, and that was why there were no captions displayed. The new requirement called for text on every slide, so I fixed it along with the above problem. Developers are responsible for changing the Sprint Backlog when they gain additional knowledge and the tasks are still directed towards the Sprint Goal (Coleman, 2024).

Communication

I had a vast majority of the project-related communication via written communication. As the Tester, I wrote to Christy:

> While writing test cases for User Stories 2, 3, and 4, I require clarification of four issues in order to set pass and fail criteria objectively. . . . Third, what should customers see when reaching the last destination?

After the wellness change, I wrote to Christy and the Tester as the Developer:

> Christy, please confirm the approved wellness destinations and the content for each slide, whether saved preferences and vacation-type filters are still in scope, and what Next should do on the final slide. . . . Could you both reply before tomorrow's Daily Scrum? I will record your answers in the backlog item and use them to plan and verify the changes.

The beginning of the message stated my purpose of communication and every question could be easily answered with a decision. The second message went further, adding the deadline

tied to the Daily Scrum event, to get the answers in time when any work was not started. Communicating with both the Product Owner and the Tester involved the testing into the change at the beginning and not at the end, and promising to log the answers in the backlog item would make everybody work from the same decision. My last-slide question originated from the test cases I had written being the Tester, which shows that communication from multiple roles helped to form a better question.

Organizational Tools

The main organizational tool for this project was the Product Backlog with all stories ordered and sized. It made it easier for us to start the Sprint Planning and adjust the order of the work after the wellness change. The user story sheets and the test case template formed a pair. All acceptance criteria corresponded to test steps, which allowed demonstrating to Amanda the fulfillment of acceptance criteria in the Sprint Review, not showing the screens. The charter stated the rules for every Scrum event, including a 15-minute Daily Scrum for discussing impediments and Sprint Retrospective with an agreed action item. The weak spot was Email, since the answers were scattered in the separate inboxes. A shared board such as Azure Boards or Jira would collect Christy's decisions in the work item and allow seeing them all at once during the Daily Scrum. Scrum requires transparency, and the inspect-and-adapt cycle would work only if everybody is looking at the same information.

Evaluation of the Agile Process

There were many things Scrum has done right for this project. Changes were absorbed without having to start over, the questions were being raised at a stage when they could be cheaply answered, and every Sprint delivered to Amanda something to be reviewed. There were drawbacks as well. There were gaps even with light stories. There were no limits set on the page

load time or the text length. The story Personalized Travel Destination List mentioned that the unlikely destinations had to be filtered or ranked low without explaining how that has to be done. The changes meant rework; any task requiring Christy's answer had to wait until she replied.

Nevertheless, even with the drawbacks mentioned above, Scrum was the best choice for the SNHU Travel project. The requirements were gathered from a focus group and were always going to shift upon seeing actual screens, as the wellness change showed. With the waterfall, the shift would happen only after the requirements have been signed off. The change would require a formal change request and a new design document creation; testing would be done after the development has been completed. Under Scrum, the shift cost us one update of the slide show and updating the set of test cases. I recommend that ChadaTech switches to Scrum and learn two lessons from this pilot project: write acceptance criteria with measurable limits, and keep all decisions on a shared board, not in emails.

References

Coleman, J. (2024, October 17). *Developers accountability in Scrum*. Scrum.org. https://www.scrum.org/resources/blog/developers-accountability-scrum

Verwijs, C., & Russo, D. (2023). A theory of Scrum team effectiveness. *ACM Transactions on Software Engineering and Methodology, 32*(3), Article 74. https://doi.org/10.1145/3571849